

P2 — The Security Tax

Paying for Authenticity in Milliseconds: A Certified Budget for Post-Quantum Signing on the UAV Bus and Link

Research proposal, RobustUAVs.ai project · Roger Nick Anaedevha · drafted 2026-08-13
Follow-on to *From Wire to Flight* (NDSS submission)

Abstract

The composition theorem of the parent manuscript bounds the mission-completion floor achievable when a residual budget Δ of undetected staleness reaches the navigation input. The theorem never asks *why* the input is stale. This proposal takes that indifference seriously: post-quantum authentication on a constrained UAV bus or command link imposes a latency that is, formally, the same quantity an adversary spends. Security hardening and the attacker therefore draw on one budget, and the certificate is what arbitrates between them. We propose to measure the latency of classical and post-quantum schemes on flight-controller-class hardware, model the fragmentation cost on DroneCAN and MAVLink 2, and push the result through the existing chain to obtain a *certified authentication budget*: the maximum per-message signing latency a platform can afford and remain airworthy.

1 The gap

Three facts collide, and nobody has put them in the same paper.

MAVLink 2 authenticates but does not encrypt. The protocol carries a 13-byte HMAC-SHA256 signature providing authentication and integrity, and offers no payload confidentiality [8]. Its adoption in practice is uneven and the community discussion of adding encryption remains open.

DroneCAN has no authentication at all. The CAN protocol that the ArduPilot and PX4 projects use to reach peripherals provides no origin authentication for bus frames [2]. The parent project already ingests 200,000 real frames of flooding, fuzzing and replay against exactly this surface [3], so the attack side is characterised and the defence side is empty.

Post-quantum migration is arriving on constrained platforms. Kyber-based group key agreement has been proposed for swarms [1], NIST-standardised lattice schemes [4] are being integrated into MAVLink-based systems, and post-quantum authentication for drone networks is under active development [7]; the broader UTM security picture is surveyed in [6].

Every one of those results reports the same metrics — signature size, key size, cycles, energy. **None reports what those milliseconds do to the aircraft.** And the arithmetic is unforgiving on a bus whose payload is 8 bytes in classical CAN and 64 in CAN-FD: a lattice signature of a few kilobytes becomes multi-frame fragmentation, which is latency, which is staleness of exactly the kind the parent manuscript’s Lemma 1 already bounds.

2 The claim

Security hardening and the adversary draw on one budget, and the composition theorem is what arbitrates between them.

Formally, the residual budget $\Delta(\theta) \leq H \min(\theta, s)$ of the parent manuscript bounds *adversarial* staleness. Authentication contributes *self-inflicted* staleness Δ_{auth} , a deterministic function of the scheme, the message rate, and the transport MTU. The certificate consumes their sum:

$$\rho(\theta) = (\Delta(\theta) + \Delta_{\text{auth}}) \gamma e^{LT_c} \leq m, \quad (1)$$

which, for a given corridor margin m , yields a **certified authentication budget**: the largest Δ_{auth} compatible with airworthiness. Three corollaries follow immediately, and all are testable.

Proposition 1 (Detector tightness buys cryptographic headroom). *Lowering θ shrinks $\Delta(\theta)$ and therefore enlarges the admissible Δ_{auth} at fixed m . Network-layer detection and cryptographic hardening trade against one another through the corridor margin.*

Proposition 2 (A margin below which nothing certifies). *For each transport there exists a corridor margin below which no post-quantum scheme in the candidate set satisfies Eq. (1) at any detector setting. The paper can name that margin per transport.*

Proposition 3 (Ranking by size is the wrong ranking). *Because fragmentation is a step function of the transport MTU rather than linear in signature bytes, the ordering of schemes under Eq. (1) need not agree with the ordering by signature size that the cryptographic literature reports.*

■ *Proposition 1 is the sentence the paper lives or dies on, and it is only interesting if the numbers are real. A modelled Δ_{auth} built from cited cycle counts would make the whole result a restatement of the inputs. The measurement is the contribution; the theorem is the frame.*

3 What is reused, what is new

Reused unchanged	New
Theorem 1 and its proof (Δ is Δ , whatever its source) Lemma 1, the residual-budget bound	Δ_{auth} : scheme $\times$ rate $\times$ MTU $\rightarrow$ seconds measured signing and verification latency on flight-controller-class hardware
certificates/engine.py and the corridor family the HCRL UAVCAN ingest, 200k real frames [3]	the budget-allocation result and Propositions 1–3 a defence side for the c2_link and intra_bus sublayers, which the schema has but the corpus does not

4 Experiment plan

4.1 Measure, do not cite

Bench Ed25519, HMAC-SHA256 (the MAVLink 2 baseline), the NIST lattice signature and KEM standards [4], and a compact-signature alternative, on two targets: a flight-controller-class microcontroller (STM32H7, as on Pixhawk 6) and a companion-computer-class board. Report sign latency, verify latency, and — the one that matters and the one nobody reports — *end-to-end frame-to-frame* latency under a realistic message mix.

4.2 Fragmentation model

Map signature size to DroneCAN frame count to bus occupancy at realistic message rates, yielding Δ_{auth} . Validate against a real bus, using the timing of the benign portion of the HCRL UAVCAN capture as the baseline.

4.3 Compose

Push Δ_{auth} through the existing chain. Produce the certified authentication budget as a function of corridor margin, and the scheme ranking that Proposition 3 predicts will differ from the size ordering.

4.4 The adversarial interaction

The most interesting case, and a genuinely new attack: an adversary who *induces* authentication work — flooding the bus with frames carrying invalid signatures — turns the defence into the delay attack. Verification cost is incurred before the signature is known to be bad, so the defender pays per frame. This is directly measurable on the HCRL surface, and the composition theorem bounds its effect on the mission.

This attack is dual-use, and the write-up must handle it the way the parent manuscript handled its evasion analysis: the mechanism is not novel to anyone who has implemented signature verification, the result bounds what the attacker achieves, and the practical output is a defensive configuration procedure. Rate-limiting and cheap pre-filters are the mitigation, and they belong in the same section as the attack.

5 Relation to the parent manuscript’s limitations

Limitation	Closed?	How
Analytical anchor and detector scope (one detector, a scalar operating point)	weakly	exercises a second attack surface, the bus, but still with a scalar operating point
Evidence-class imbalance	no	unchanged

P2’s value is not limitation-closing; it is that it adds a defence side to two sublayers the schema already models and the corpus already covers, and it does so with data that needs no dataset gatekeeper.

6 Platform integration

A **Crypto Budget** page: choose a transport (MAVLink 2, DroneCAN, CAN-FD), a scheme, a message rate, and a corridor margin; receive the certified floor and a pass/fail against the $MCR \geq 0.90$ reference consistent with DO-326A [5]. This is the most operationally useful page the platform could carry, and it requires no dataset at all.

7 Risks

- **Hardware access.** A microcontroller development board is inexpensive; a full flight controller is needed only for the bus validation, not for the timing measurement. If neither is reachable, benchmarks on a pinned ARM target are a defensible second best — but the paper must say so, and the provenance tag must say so too.
- **Reviewers may read it as post-quantum benchmarking.** The framing must lead with the budget result, not the numbers. The numbers are the instrument, not the finding.
- **Scheme selection may date quickly.** Pin the candidate set to the standardised algorithms [4] and report the model, so a new scheme can be placed in the ranking without re-running the campaign.

8 Venue and timeline

Target ACM AsiaCCS, ESORICS, or IEEE EuroS&P; NDSS if the induced-authentication attack lands cleanly. Four to five months, and the bench work parallelises with P1’s measurement campaign.

References

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- [2] DroneCAN. DroneCAN Specification: MAVLink Bus Transport Layer. protocol specification. accessed 2026-08-13.
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- [4] National Institute of Standards and Technology. Module-Lattice-Based Key-Encapsulation Mechanism Standard (FIPS 203); Module-Lattice-Based Digital Signature Standard (FIPS 204); Stateless Hash-Based Digital Signature Standard (FIPS 205). Technical report, NIST, 2024.
- [5] RTCA and EUROCAE. DO-326A / ED-202A: Airworthiness Security Process Specification. Technical report, RTCA, Inc. / EUROCAE, 2014.

- [6] see arXiv record. A Survey of Security Challenges and Solutions for UAS Traffic Management (UTM) and Small Unmanned Aerial Systems. arXiv:2601.08229, 2026. author list to be confirmed at citation time.
- [7] see IEEE Xplore record. Post-Quantum Authentication and Communication Security for Drone Networks. *IEEE Transactions on Quantum Engineering*, 2025. early access; volume, pages and author list to be confirmed.
- [8] Trout Software. Securing UAV Ground Stations: MAVLink Vulnerabilities. industry analysis. accessed 2026-08-13; MAVLink 2 signs but does not encrypt.